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RESEARCH ARTICLE

Genome-wide identification, characterization, and expression analysis of the *SWEET* gene family in cucumber



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Abstract

SWEETs (sugars will eventually be exported transporters) are a novel class of recently identified sugar transporters that play important roles in diverse physiological processes. However, only a few species of the plant *SWEET* gene family have been functionally identified. Up till now, there has been no systematic analysis of the *SWEET* gene family in Cucurbitaceae crops. Here, a genome-wide characterization of this family was conducted in cucumber (*Cucumis sativus* L.). A total of 17 *CsSWEET* genes were identified, which are not evenly distributed over the seven cucumber chromosomes. Cucumber *SWEET* protein sequences possess seven conserved domains and two putative serine phosphorylation sites. The phylogenetic tree of the *SWEET* genes in cucumber, *Arabidopsis thaliana*, and *Oryza sativa* was constructed, and all the *SWEET* genes were divided into four clades. In addition, a number of putative *cis*-elements were identified in the promoter regions of these *CsSWEET* genes: nine types involved in phytohormone responses and eight types involved in stress responses. Moreover, the transcript levels of *CsSWEET* genes were analyzed in various tissues using quantitative real-time polymerase chain reaction. A majority (70.58%) of the *CsSWEET* genes were confined to reproductive tissue development. Finally, 18 putative watermelon *ClSWEET* genes and 18 melon *CmSWEET* genes were identified that showed a high degree of similarity with *CsSWEET* genes. The results from this study provided a basic understanding of the *CsSWEET* genes and may also facilitate future research to elucidate the function of *SWEET* genes in cucumber and other Cucurbitaceae crops.

Keywords: cucumber, gene expression, phylogenetic analysis, sugar transporter, *SWEET*, watermelon

1. Introduction

Soluble sugars, which are the primary products of photosynthesis, can be used as sources of carbon skeletons for the biosynthesis of many other cellular compounds, signals, osmolytes, and transport molecules, and they can be used as transient energy storage (Chen *et al.* 2015a). In plants, sugars are synthesized in source leaves during photosynthesis and are eventually delivered to various heterotrophic tissues such as roots, seeds, and fruits to enable their

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growth and development (Lalonde *et al.* 2004; Chen *et al.* 2014; Ruan 2014). Not surprisingly, this cellular exchange of sugars requires transport proteins to mediate uptake or release from cells or subcellular compartments (Chen *et al.* 2015a). Thus far, three families of eukaryotic sugar transporters have been identified as key players, namely, monosaccharide transporters (MSTs), sucrose transporters (SUTs), and *SWEETs* (sugars will eventually be exported transporters) (Eom *et al.* 2015). It is now widely accepted that these sugar transporters play key roles in the entire plant sugar translocation and allocation, and they significantly determine the crop output and quality (Kühn and Grof 2010; Ayre 2011; Slewinski 2011; Chen *et al.* 2015a; Eom *et al.* 2015).

MSTs and SUTs belong to the major facilitator superfamily and are characterized by 12 transmembrane domains (Kühn and Grof 2010; Ayre 2011; Slewinski 2011). *SWEETs* are a recently identified family of sugar efflux transporters that selectively transport monosaccharides or disaccharides across intracellular or plasma membranes, and they belong to the MtN3/saliva family (Pfam code PF03083) (Chen *et al.* 2010; Xuan *et al.* 2013; Chen 2014; Eom *et al.* 2015). *SWEETs* are widely distributed in various organisms, including prokaryotes, animals, and plants (Yuan and Wang 2013; Patil *et al.* 2015). It has been predicted that eukaryotic *SWEET* proteins typically comprise seven α -helical transmembrane (TM) domains that are organized as tandem repeats of two 3-TM domains (containing two conserved MtN3/saliva motifs) that are separated by a single TM that is less conserved (Xuan *et al.* 2013). The resulting structure has been described as the 3-1-3 TM *SWEET* structure (Chen *et al.* 2010). In contrast to eukaryotic *SWEET* proteins, the prokaryotic *SWEET* proteins, which are known as Semi*SWEETs*, contain only one 3-TM, possibly indicating that eukaryotic *SWEETs* have evolved *via* duplication and fusion of the basic 3-TM unit that is present in prokaryotic Semi*SWEETs* (Xuan *et al.* 2013).

In plants, a growing body of evidence suggests that *SWEETs* are involved in many significant physiological and biochemical processes where sugar efflux plays a vital role, such as phloem loading for long-distance sugar transport, nectar secretion, pollen nutriture, seed filling, plant-pathogen interaction, and biotic and abiotic stress regulation (Chen 2014; Chandran 2015; Chen *et al.* 2015a; Eom *et al.* 2015). In *Arabidopsis* (*Arabidopsis thaliana*), *AtSWEET11* and *AtSWEET12*, which localize to the plasma membrane of phloem parenchyma cells and export sucrose from these cells into the apoplast, are responsible for the first step of phloem loading of sucrose in long-distance transport (Chen *et al.* 2012). The nectar-specific *AtSWEET9* is essential for nectar production and can function as a sucrose efflux transporter (Lin *et al.* 2014). Similarly, the *NEC1* gene

isolated from *Petunia* is thought to play a role in nectar secretion because it is highly expressed in nectar cells (Ge *et al.* 2000, 2001). Additionally, many *SWEET* genes, such as *AtSWEET5* (also called *VEX1*), which is expressed at different stages of pollen development (Engel *et al.* 2005), *AtSWEET8* (also called *RPG1*), which is strongly expressed in microspores and tapetum during male meiosis (Guan *et al.* 2008), and *AtSWEET13* (also called *RPG2*), which is highly expressed in anthers (Sun *et al.* 2013), play a broader role in pollen nutriture. In rice (*Oryza sativa*), *OsSWEET5* is also expressed in anthers (Zhou *et al.* 2014). *SWEETs* were also shown to be involved in seed filling. *AtSWEET11*, *AtSWEET12*, and *AtSWEET15* (also called *SAG29*) exhibit specific spatiotemporal expression patterns in developing seeds of *Arabidopsis*, but only the triple knockout mutant showed severe seed defects, including retarded embryo development, reduced seed weight, and lower starch and lipid content, causing a wrinkled seed phenotype at maturity (Chen *et al.* 2015b). In contrast with *Arabidopsis*, seed filling in domesticated maize (*Zea mays*) and rice depends on hexose transport across the basal endosperm transfer layer. This process is controlled by *ZmSWEET4c* in maize and *OsSWEET4* in rice (Sosso *et al.* 2015). Moreover, *SWEETs* are the targets of extracellular pathogens that modify the expression of *SWEETs* to gain sugars to fuel their own growth and reproduction (Yuan *et al.* 2009; Chen *et al.* 2010; Streubel *et al.* 2013; Chong *et al.* 2014; Cohn *et al.* 2014; Hu *et al.* 2014; Chen *et al.* 2015). *SWEETs* may also play important roles in biotic and abiotic stress tolerance. For example, *AtSWEET15* is obviously induced during leaf senescence and osmotic stresses, including high salinity, cold, and drought, *via* an abscisic acid-dependent pathway (Seo *et al.* 2011). The *AtSWEET15*-overexpressing transgenic plants display an accelerated leaf senescence and are hypersensitive to high salinity stress, whereas the *atsweet15*-deficient mutant lines are less sensitive to high salinity (Seo *et al.* 2011). Studies have revealed that *AtSWEET16* and *AtSWEET17* are the primary factors in controlling glucose or fructose content in the tonoplasts of *Arabidopsis* leaves and roots, especially under cold or low nitrogen supply conditions (Chardon *et al.* 2013; Klemens *et al.* 2013; Guo *et al.* 2014). More recently, studies have shown that *AtSWEET11* and *AtSWEET12* are also involved in cold stress or water deficit conditions (Hir *et al.* 2015; Durand *et al.* 2016). The *AtSWEET11;12* double mutant exhibited greater freezing tolerance than the wild-type and both single mutants (Hir *et al.* 2015). Additionally, the transcript levels of *AtSWEET11*, *AtSWEET12*, and *AtSUC2*, which are all involved in sucrose phloem loading, are largely up-regulated in source leaves under water deficit condition (Durand *et al.* 2016). This is in agreement with the

increase in carbon export from the leaves to the roots, which maintains an efficient root system and sustains their development (Durand *et al.* 2016).

An increasing number of studies have demonstrated that extensive functional divergence may be present within the plant *SWEET* genes. Studies on *SWEET*s in several plant species, especially *Arabidopsis* and rice, have provided a better understanding of this gene family. However, information on *SWEET* genes from more species still requires further study. To our knowledge, no systematic investigations on the *SWEET* gene family of Cucurbitaceae crops have been reported to date. Cucumber (*Cucumis sativus* L.) is one of the economically and nutritionally important vegetable crops cultivated worldwide. It belongs to Cucurbitaceae, which includes several other important species, such as watermelon (*Citrullus lanatus*) and melon (*Cucumis melo*). In addition, the sequencing of the cucumber genome (Huang *et al.* 2009) has facilitated the study of the members of the *SWEET* gene family in cucumber. In this study, we present a detailed analysis of the phylogeny, chromosome distribution, gene structure, domain architecture, *cis*-acting regulatory elements, and expression analyses of the cucumber *SWEET* genes. The putative watermelon *ClaSWEET* genes and melon *CmSWEET* genes were also identified to provide more information about the *SWEET* gene families in cucurbits. The data from this study will provide valuable information for further exploration into the functions of this significant gene family in cucumber and other Cucurbitaceae crops.

2. Materials and methods

2.1. Identification of cucumber *SWEET* genes

The method used to identify all putative *SWEET* family genes in cucumber was similar to that used for soybean (*Glycine max*) and tomato (*Solanum lycopersicum*) (Feng *et al.* 2015; Patil *et al.* 2015). Candidate genes were identified by performing a BLASTP analysis against the cucumber genome database (<http://www.icugi.org/cgi-bin/ICuGI/genome/index.cgi?organism=cucumber>). In total, the amino acid sequences of 17 *AtSWEET* genes and 21 *OsSWEET* genes those were obtained from the *Arabidopsis* Information Resource (<http://arabidopsis.org>) and NCBI (<http://www.ncbi.nlm.nih.gov>) were used as query sequences. A further search was performed *via* Hidden Markov Model (HMM) analysis using the HMMER 3.0 Program. The HMMER hits that contained the MtN3/saliva domain were compared to the BLASTP results and parsed *via* manual editing. Finally, all of the *SWEET* protein sequences of the cucumber were submitted to the Pfam database (<http://pfam.xfam.org>) (Finn *et al.* 2016), the SMART database ([delberg.de\) \(Schultz *et al.* 1998; Letunic *et al.* 2015\), and NCBI's Conserved Domain Database \(<http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>\) to eliminate genes that did not contain the known conserved domains and motifs of the MtN3/saliva/*SWEET* family members. The *SWEET* gene family members of watermelon and melon were detected by the same method from the watermelon genome database \(<http://www.icugi.org/cgi-bin/ICuGI/genome/index.cgi?organism=watermelon>\) and the melon genome database \(<https://melonomics.net>\).](http://smart.embl-hei-</p>
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The 'Compute pI/Mw' tool from ExPASy (http://web.expasy.org/compute_pi/) was used to predict the molecular weights and isoelectric points of the deduced *SWEET* proteins of cucumber that were identified from the currently available genomic database. Subcellular localization prediction of each gene was predicated by WoLF PSORT (http://www.genscript.com/psort/wolf_psort.html). The number of TM domains was predicted using the TMHMM Server v. 2.0 (<http://www.cbs.dtu.dk/services/TMHMM>).

2.2. Exon-intron structure and genome distribution

For exon-intron structure analysis, the DNA and cDNA sequences corresponding to each predicted gene from the cucumber genome database were downloaded, and then the exon-intron distribution pattern and intron phases were analyzed using the web-based bioinformatics tool Gene Structure Display Server (GSDS, <http://gsds.cbi.pku.edu.cn>) (Hu *et al.* 2015). To obtain information on the locations of cucumber *SWEET* genes, a map detailing the distribution of *CsSWEET* family members throughout the cucumber genome was drawn using the MapInspect tool. Tandem-duplicated genes were analyzed based on the method provided by Hu and Liu (2012).

2.3. Multiple sequence alignment, conserved motif prediction, and phylogenetic analysis

Multiple sequence alignments of the amino acid sequences of cucumber *SWEET* family proteins were generated using Clustal Omega (<http://www.ebi.ac.uk/Tools/msa/clustalo>) with the default parameters. Then, conserved or similar amino acid sequences were highlighted with BOXSHADE (http://www.ch.embnet.org/software/BOX_form.html). The online MEME tool (<http://meme-suite.org/tools/meme>) was later used to identify the conserved motifs of the *SWEET* protein sequences, and the generated motifs were visualized using Logo (<http://weblogo.berkeley.edu/logo.cgi>). Phylogenetic analysis was performed based on the *SWEET* protein sequences using the MEGA 5.05 Program *via* the neighbor joining (NJ) method, and a bootstrap test was performed with 1 000 interactions (Tamura *et al.* 2011).

2.4. *cis*-Element analysis of putative promoter regions

To investigate the *cis*-elements in the promoter regions of all of the obtained genes, we downloaded 1.5 kb of the genomic DNA sequences upstream of the ATG start codon corresponding to each gene from the cucumber database. The putative *cis*-regulatory elements in the promoter sequences were analyzed via the PlantCARE database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html>) (Lescot *et al.* 2002).

2.5. Plant materials, total RNA extraction, and quantitative real-time expression analysis

Cucumber (*C. sativus* L. cv. C49) plants were grown under glasshouse conditions from the end of August to December, 2015. The roots, stems, leaves, male flowers (on the day of anthesis), fruits (6 days after anthesis), and tendrils were collected from 3-mon-old plants. All samples were immediately frozen in liquid nitrogen and stored at -80°C until use.

Total RNAs were extracted using the Column Plant RNAout Kit (Tiandz, China) according to the manufacturer's instructions. After removal of the potential genomic DNA with DNAase I (Tiangen, China), the DNA-free RNAs were reverse transcribed into a single strand of cDNA, which was used as the template for real-time PCR analysis. The gene specific primers used for analysis are listed in Appendix A. The PCR primers were designed to avoid the conserved region, and the specificity of each primer to their target genes was verified using the BLASTN Program of the cucumber genome database. Quantitative real-time PCR was performed with SYBR Green I Master (Roche, Switzerland) using a LightCycler[®] 480 system (Roche). Three biological replicates and three technical replicates were performed for each reverse transcription-PCR procedure. Amplification was initiated with a denaturation step of 5 min at 95°C , followed by 45 cycles of 95°C for 10 s, 58°C for 30 s, and 72°C for 30 s. The specificity of the reactions was verified via melting curve analysis. A cucumber *actin* gene (*Csa6M484600.1*) was detected as an internal control to calibrate the relative expression. The $2^{-\Delta\Delta\text{C}_T}$ method was used to calculate the relative expression level of the target gene (Livak and Schmittgen 2001).

3. Results

3.1. Identification of cucumber *SWEET* gene family members and their distribution on chromosomes

The *SWEET* genes were identified based on BLASTP searches against the cucumber genome database. A total

of 20 *CsSWEET* genes with confirmed conserved domains were identified from the cucumber genome. However, the nucleotide sequence of *Csa2M083730.2* was completely identical to that of *Csa2M083730.1*. In addition, there are two putative gene models of *Csa6M343690* (*Csa6M343690.1* and *Csa6M343690.2*) and *Csa1M051710* (*Csa1M051710.1* and *Csa1M051710.2*) in the cucumber genome database that differ in the predicted splice sites. Therefore, *Csa6M343690.2* and *Csa1M051710.2*, whose nucleotide sequences are shorter than those of *Csa6M343690.1* and *Csa1M051710.1*, respectively, as well as *Csa2M083730.2*, were excluded in the following analysis. The remaining 17 *CsSWEET* genes were named according to their homologous genes in *Arabidopsis* (Table 1). Each *AtSWEET* gene corresponded to approximately one to three *CsSWEET* genes (Table 1). The open reading frames (ORFs) of the *CsSWEET* genes ranged from 687 to 888 bp in length and encoded polypeptides from 228 to 295 aa in length (Table 1). The molecular weights (kDa) and isoelectric points (pI) of these *CsSWEET*-deduced polypeptides ranged from 25.80 to 33.05 kDa, and from 7.72 to 9.58, respectively (Table 1). Moreover, the *CsSWEET* genes varied in subcellular localization, which included the chloroplast, vacuole, nucleus, plasma membrane, and extracellular (Table 1).

The chromosomal location of each *CsSWEET* gene was determined based on the genomic sequences of cucumbers. A total of 17 *CsSWEET* genes were localized on six of the seven chromosomes (Fig. 1). Chromosome 1 contained the highest number of *CsSWEET* genes (six, 35.3%), whereas no *CsSWEET* genes were found on chromosome 7 (Fig. 1). Chromosome 3 contained three *CsSWEET* genes (17.6%), and each of the remaining chromosomes contained two *CsSWEET* genes (11.8%). A total of nine *CsSWEET* genes (*CsSWEET 1*, *5a–5c*, *10*, *12a*, *15*, *17a*, and *17b*) were localized in the 5' to 3' directions on the chromosomes in cucumber, whereas the remaining eight genes were in the opposite direction (Fig. 1, Table 1). Two pairs of tandem-duplicated genes (*CsSWEET5b* and *CsSWEET5c*; *CsSWEET12b* and *CsSWEET12c*) were found to be distributed on cucumber chromosomes (Fig. 1).

3.2. Conserved domain analysis of *SWEET* gene family members in cucumber

The typical plant *SWEET* protein contains seven TM domain helices that consist of two tandem repeats of 3-TM units separated by a single TM unit (Chen *et al.* 2010; Xuan *et al.* 2013). We deduced the amino acid sequences of the 17 members of the cucumber *SWEET* gene family and submitted them to the TMHMM Server v. 2.0 in FASTA format using the default values. The results showed that 14

Table 1 Characteristics of the SWEET genes in cucumber

Gene name	Gene ID	Location	ORF length (bp)	TMD ¹⁾	Subcellular localization ²⁾	Protein length (aa)	Molecular weight ³⁾ (kDa)	pI ³⁾	Arabidopsis homologous
CsSWEET1	Csa6M343690.1	15558386–15561811 (+ strand)	759	7	CH/VA	252	27.72	9.13	AtSWEET1/AT1G21460
CsSWEET2	Csa4M622870.1	20033119–20034999 (– strand)	702	7	NU	233	26.07	9.18	AtSWEET2/AT3G14770
CsSWEET3	Csa5M139600.1	3654165–3656586 (– strand)	780	7	PM	259	29.14	9.38	AtSWEET3/AT5G53190
CsSWEET5a	Csa1M046010.1	5246445–5247639 (+ strand)	717	7	EX	238	26.93	9.18	AtSWEET5/AT5G62850/VEX1
CsSWEET5b	Csa3M636410.1	33355218–33357089 (+ strand)	711	6	PM	236	26.27	8.72	AtSWEET5/AT5G62850/VEX1
CsSWEET5c	Csa3M636420.1	33364868–33369110 (+ strand)	744	7	PM	247	28.00	8.63	AtSWEET5/AT5G62850/VEX1
CsSWEET7a	Csa4M054810.1	4561080–4563937 (– strand)	786	7	PM	261	28.92	9.30	AtSWEET7/AT4G10850
CsSWEET7b	Csa5M148830.1	4455565–4457775 (– strand)	798	7	PM	265	29.15	9.58	AtSWEET7/AT4G10850
CsSWEET9	Csa2M083730.1	6768446–6772168 (– strand)	789	7	CH	262	29.89	8.94	AtSWEET9/AT2G39060
CsSWEET10	Csa1M001300.1	165089–166577 (+ strand)	879	7	CH	292	32.75	9.23	AtSWEET10/AT5G50790
CsSWEET12a	Csa1M001290.1	158986–160345 (+ strand)	828	6	CH	275	30.92	8.42	AtSWEET12/AT5G23660
CsSWEET12b	Csa1M051710.1	6084560–6086365 (– strand)	888	7	CH	295	33.05	8.90	AtSWEET12/AT5G23660
CsSWEET12c	Csa1M051720.1	6094146–6095586 (– strand)	876	7	CH	291	32.88	7.72	AtSWEET12/AT5G23660
CsSWEET15	Csa6M092510.1	6333485–6336129 (+ strand)	834	7	CH	277	31.43	9.06	AtSWEET15/AT5G13170/SAG29
CsSWEET17a	Csa1M569280.1	20889484–20895553 (+ strand)	888	7	CH	295	32.41	9.39	AtSWEET17/AT4G15920
CsSWEET17b	Csa2M031160.1	3050146–3052006 (+ strand)	687	7	VA	228	25.80	9.05	AtSWEET17/AT4G15920
CsSWEET17c	Csa3M159450.1	10789065–10792612 (– strand)	735	6	CH	244	27.03	8.85	AtSWEET17/AT4G15920

¹⁾ The number of transmembrane domains (TMD) was predicted by TMHMM Server v. 2.0.

²⁾ The subcellular localizations were predicted by WoLFPSORT. CH, chloroplast; VA, vacuole; NU, nucleus; PM, plasma membrane; EX, extracellular.

³⁾ The molecular weight and theoretical isoelectric point (pI) were predicted using 'Compute pI/Mw' tool from ExPASy.

CsSWEET proteins had 7 TMs, and the rest (CsSWEET5b, CsSWEET12a, and CsSWEET17c) had 6 TMs (Table 1; Appendix B). In detail, CsSWEET5b and CsSWEET17c lacked the TM2 domain, whereas CsSWEET12a lacked the TM7 domain (Appendix B). In soybean, there are also some GmSWEET proteins that have 6 TMs, not 7 TMs (Patil *et al.* 2015). The TM domain sequence logo was generated *via* an online Logo tool to highlight the amino acid residues conserved in the seven TM domains of cucumber SWEET proteins (Fig. 2).

Sequence alignment revealed that the 7 TMs were relatively conserved (Fig. 3). Furthermore, there were 2 possible serine phosphorylation sites located on the inner side of the membrane area. The first serine phosphorylation site was located between TM1 and TM2 (Fig. 3). However, 2 proteins (CsSWEET2 and CsSWEET9) contained threonine instead of serine, while CsSWEET5a contained alanine (Fig. 3). All of these amino acids can be phosphorylated except for alanine. The second serine phosphorylation site was located between TM5 and TM6 (Fig. 3). All of the CsSWEET proteins contained this site (Fig. 3).

3.3. Phylogenetic and structural analysis of SWEET genes in cucumber

Arabidopsis and rice, which have 17 and 21 SWEET gene members (Chen *et al.* 2010), respectively, are model plants. In addition, the functions of several *AtSWEET* genes and *OsSWEET* genes have been characterized (Chen 2014; Chandran 2015; Chen *et al.* 2015a; Eom *et al.* 2015). To better understand the evolutionary origin and putative function of the cucumber SWEET genes, an unrooted phylogenetic tree was constructed by aligning 17 CsSWEET protein sequences, 17 *AtSWEET* protein sequences, and 21 *OsSWEET* protein sequences using the Clustal Omega and MEGA 5.05 software (Fig. 4). These SWEET protein sequences fall into four clades (Fig. 4) based on the classification by Chen *et al.* (2010). The numbers of SWEET genes in the four different clades of cucumber, *Arabidopsis*, and rice were compared in Appendix C. Each clade contains CsSWEET, *AtSWEET*, and *OsSWEET* (Appendix C). In detail, clade I contains three CsSWEETs (CsSWEET1–3), three *AtSWEET*s (AtSWEET1–3), and six *OsSWEET*s (*OsSWEET*1a, 1b, 2a, 2b, 3a, and 3b); Clade II contains five CsSWEETs (CsSWEET 5a–5c, 7a, and 7b), five *AtSWEET*s (AtSWEET4–8), and nine *OsSWEET*s (*OsSWEET*4, 5, 6a, 6b, and 7a–7e); Clade III contains six CsSWEETs (CsSWEET9, 10,

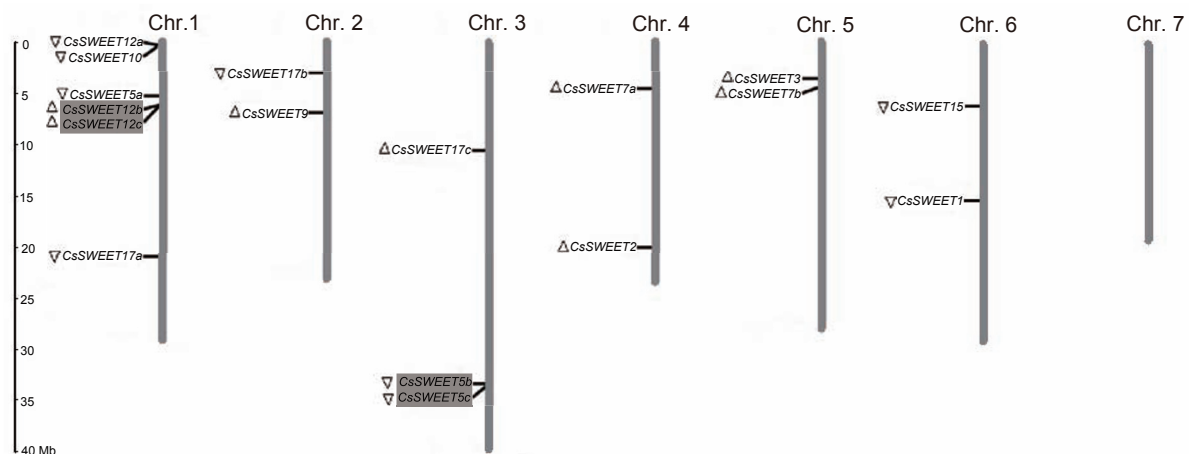


Fig. 1 Distribution of 17 cucumber *SWEET* genes on chromosomes. The bars indicate the seven chromosomes of cucumbers and the chromosome numbers are demonstrated at the top of each chromosome. The scale ruler on the left side shows the physical distance of the chromosomes. The relative positions of *CsSWEETs* are marked on the chromosomes. The arrow-heads next to gene names show the direction of transcription. Tandem-duplicated genes are indicated by the gray rectangles.

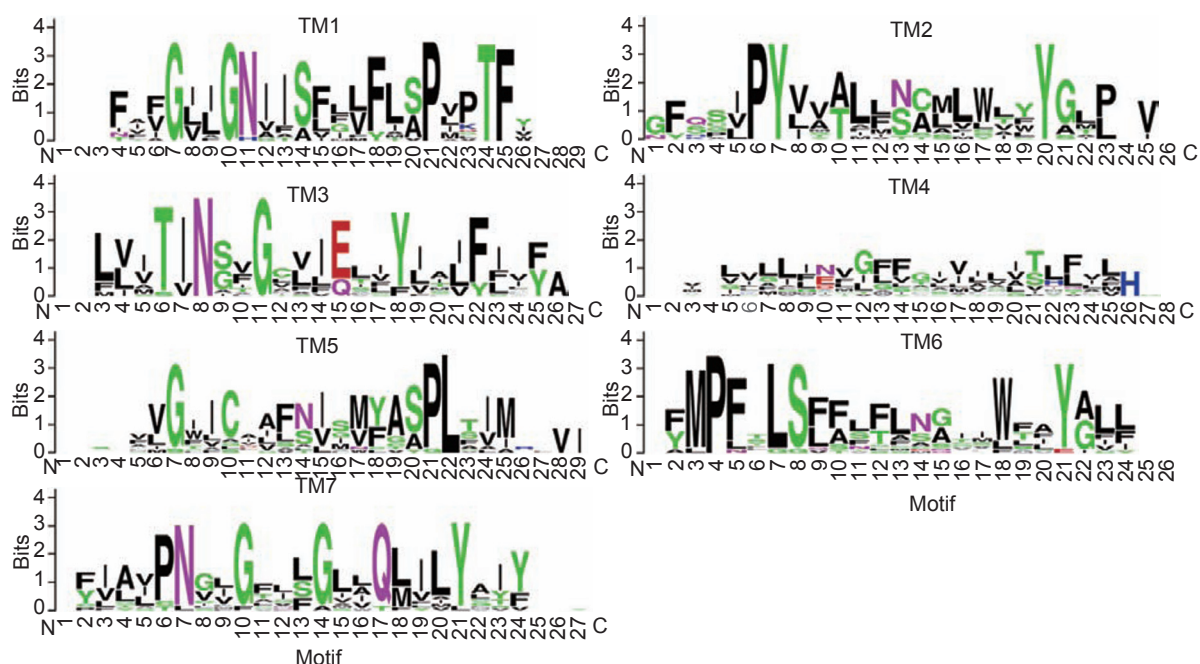


Fig. 2 Seven transmembrane (TM) sequence logos for *CsSWEET* proteins using the MEME Program. TM motifs are displayed by stacks of letters at each site. The total height of the stack is the “information content” of that site in the motif in bits. The height of each letter in a stack is the probability of the letter at that site multiplied by the total information content of the stack.

12a–12c, and 15), seven *AtSWEETs* (*AtSWEET9*–15), and five *OsSWEETs* (*OsSWEET11*–15); Clade IV contains three *CsSWEETs* (*CsSWEET17a*–17c), two *AtSWEETs* (*AtSWEET16* and 17), and one *OsSWEET* (*OsSWEET16*) (Fig. 4).

Analysis of the gene structure can provide valuable information on both the exon-intron landscape and the phylogenetic relationships within the *CsSWEET* gene family.

Based on the constructed phylogenetic tree, the exon-intron structures and intron phases of the 17 *CsSWEET* genes were analyzed using the online GSDS tool with their full-length CDS sequences and corresponding genomic DNA sequences (Fig. 5). The pattern of exon-intron classification was consistent with the phylogenetic tree (Fig. 5). In the cucumber *SWEET* gene family, all *CsSWEET* genes contained introns in their genomic sequences. *CsSWEET1*–3, 5a–5c,

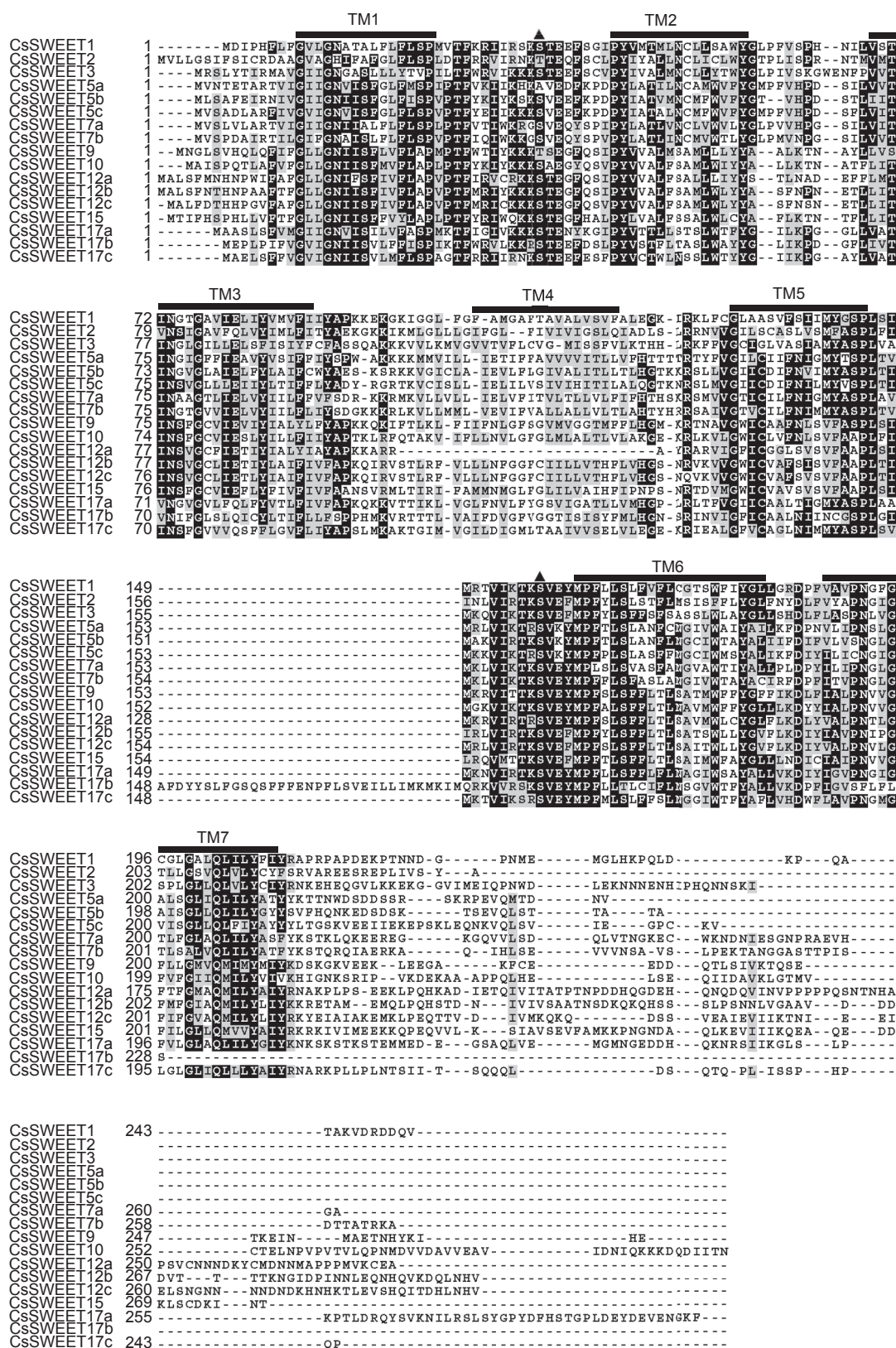


Fig. 3 Multiple sequence alignment of the cucumber *SWEET* gene family. The positions of the seven transmembrane (TM) domains (TM1 to TM7) are indicated by lines above the sequences. Triangle, conserved serine predicted to be a phosphorylation site by NetPhos 2.0 (<http://www.cbs.dtu.dk/services/NetPhos>). Clustal Omega was used to perform a homologous sequence alignment of the amino acid sequences of 17 cucumber *SWEET* family proteins, and BOXSHADE was used to generate the diagram.

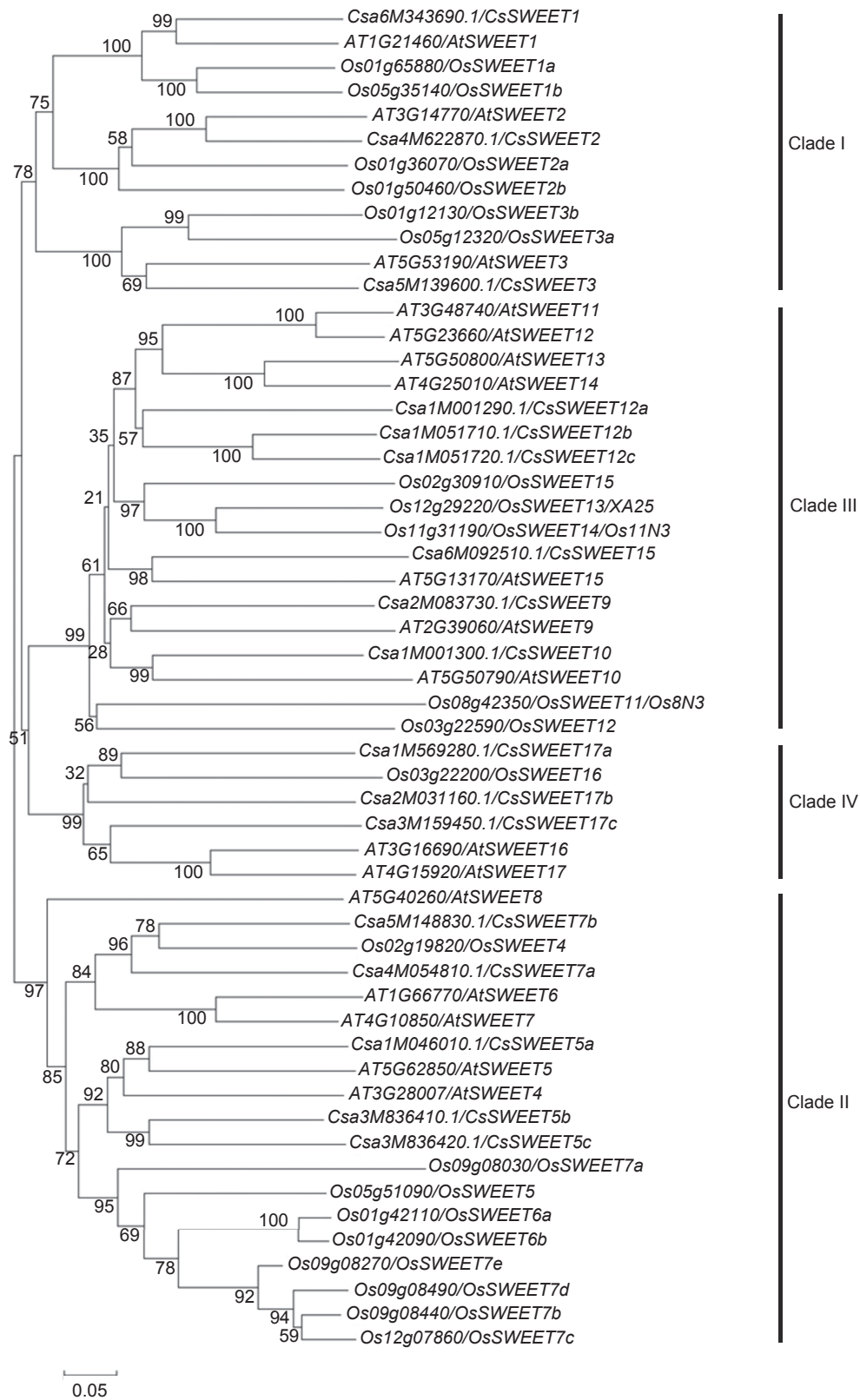


Fig. 4 Phylogenetic relationships of *SWEETs* genes in cucumber, *Arabidopsis*, and rice. The unrooted phylogenetic tree was generated based on the amino acid sequences by the neighbor joining (NJ) method using MEGA 5.05. Bootstrap supports from 1000 replicates are indicated at each branch. *SWEETs* genes fall into four clades that are labeled as I, II, III, and IV. At, *Arabidopsis*; Cs, cucumber; Os, rice.

9, 12b, 12c, 15, 17a, and 17c contained six exons, and their intron phase patterns were similar with the exception of *CsSWEET1* (Fig. 5). The remaining gene family members all contained five exons and exhibited similar intron phase patterns (Fig. 5).

3.4. *cis*-Acting regulatory elements in *SWEET* gene promoters in cucumber

To understand the transcriptional control of the *CsSWEETs*, sequences of the 1.5 kb upstream regions from the translation initiation site (ATG) were extracted for the 17 genes except for *CsSWEET17a*, for which a 1 058-bp promoter sequence was obtained because of the shorter intergenic space between *CsSWEET17a* and another gene. The *cis*-elements in the upstream sequences of the 17 genes were subjected to analysis on the PlantCARE server.

Notably, nine types of phytohormone-responsive *cis*-elements were identified in the promoter regions of *CsSWEETs*, including one abscisic acid-responsive element (ABRE), two methyl jasmonate-responsive elements (CGTCA-motif and TGACG-motif), one ethylene-responsive element (ERE), three gibberellin-responsive elements (GARE, P-box, and TATC-box), one salicylic acid-responsive element (TCA-element), and one auxin-responsive element (TGA-element) (Table 2). The promoters of the 17 *CsSWEET* genes contained at least one phytohormone-responsive *cis*-element with the exception of *CsSWEET7a* (Table 2). *CsSWEET10* and *CsSWEET17b* contained the highest number (five) of phytohormone-responsive elements (Table 2). In addition to the regulation by phytohormone-responsive *cis*-elements, *CsSWEET* genes may also respond to various environmental stresses, such as anaerobic, heat,

low temperature, drought, wounding, fungal, and circadian (Table 2). The anaerobic-responsive element (ARE) was present in ten *CsSWEET* genes (Table 2), suggesting important roles for these genes in anaerobic stress responses. However, the low temperature-responsive element (LTR) and the wound-responsive element (WUN-motif) were present only in *CsSWEET12b* and *CsSWEET5b*, respectively (Table 2). These data indicated that *CsSWEETs* may participate in environmental stresses via a complex mechanism and that each *CsSWEET* gene could be induced by different environmental stimuli.

3.5. Expression profiles of *CsSWEET* genes in different tissues or organs of the cucumber

To gain insights into the temporal and spatial transcription patterns and putative functions of *CsSWEET* genes during growth and development, real-time PCR was performed to analyze the transcription levels in various tissues or organs, including the roots, stems, leaves, male flowers, fruits, and tendrils. Although the expression of most of the 17 *CsSWEET* genes was detected in most selected organs, their expression levels varied considerably (Fig. 6). For example, *CsSWEET2* and *CsSWEET17c* were mainly expressed in cucumber tendrils (Fig. 6-B and Q). The transcript abundances of *CsSWEET3* and *CsSWEET15* were higher in the stems than in any other tissues (Fig. 6-C and N). Twelve genes (*CsSWEET1*, 5a-5c, 7a, 7b, 9, 10, 12a, 12c, 17a, and 17b) (70.58%) may have crucial functions in male flowers because of their higher expression levels than in other organs (Fig. 6-A, D–K, M, O, and P). Notably, the transcript abundance of *CsSWEET12b* was considerably higher in the stems and male flowers (Fig. 6-L). Additionally, three genes

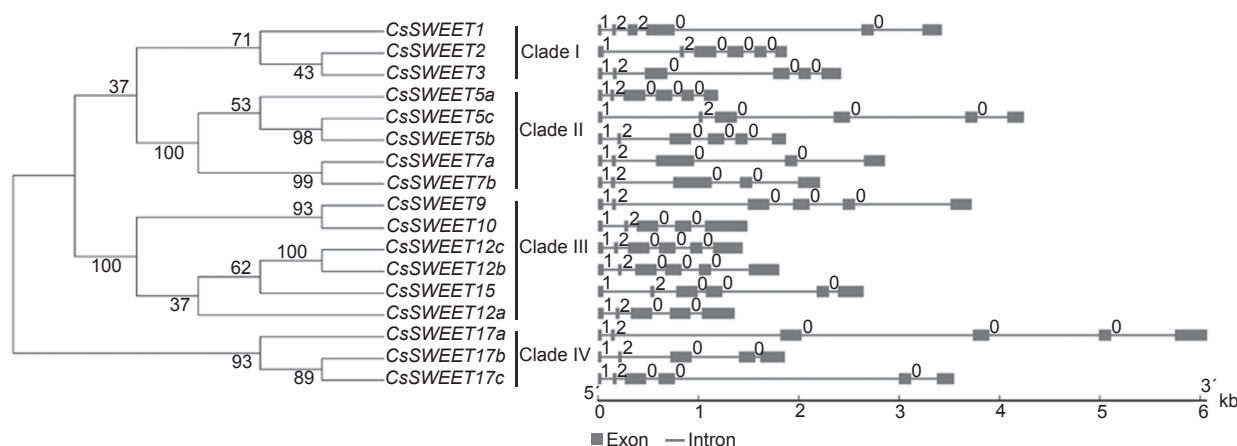


Fig. 5 Phylogenetic analysis and exon-intron structures of 17 *SWEET* genes identified in cucumbers. 0 indicates an intron located between two codons. 1 indicates an intron inserted at the first base of a codon. 2 indicates an intron inserted at the second base of a codon.

Table 2 Types and numbers of known phytohormone- and stress-related elements in the upstream regions of CsSWEET genes¹⁾

Gene name	cis-Acting regulatory element																Total	
	ABRE	CGTCA- motif	TGACG- motif	ERE	GARE	P-box	TATC- box	TC-A- element	TGA- element	ARE	HSE	LTR	MBS	TC-rich repeats	WUN- motif	Box- W1		Circadian
CsSWEET1	1				1			1		3			1	1				8
CsSWEET2					1			1	1		2			3			1	9
CsSWEET3				1		2	1			1			1	1				7
CsSWEET5a				1	1	1		1		2								5
CsSWEET5b		1												2	1			4
CsSWEET5c					1	1		1			1			1				5
CsSWEET7a											2			2		1		5
CsSWEET7b					1					1			1			1		4
CsSWEET9					1			1	1	3				1			1	8
CsSWEET10			1	1	2			1		1			1			1		8
CsSWEET12a						1		1		1			1					4
CsSWEET12b					1			1			1	1		1				5
CsSWEET12c	1									1	2			1				5
CsSWEET15	1			1				1		1						1		5
CsSWEET17a						1				1				1				3
CsSWEET17b	1		1	1				1	1				1	2			3	11
CsSWEET17c	2		1								1			2		1		7

¹⁾ ABRE, abscisic acid-responsive element; ERE, ethylene-responsive element; GARE, gibberellin-responsive element; GARE, anaerobic-responsive element; HSE, heat shock-responsive element; LTR, low temperature-responsive element; MBS, MYB binding site; ARE, TGA-element, and TGA-element are involved in the abscisic acid responsiveness, ethylene responsiveness, salicylic acid responsiveness, and auxin responsiveness, respectively. CGTCA-motif and TGACG-motif are involved in the methyl jasmonate responsiveness. GARE, P-box, and TATC-box are involved in the gibberellin responsiveness. ARE, HSE, LTR, MBS, WUN-motif, and Box-W1 are involved in the anaerobic induction, heat stress responsiveness, low temperature responsiveness, drought inducibility, wound responsiveness, and fungal responsiveness, respectively. TC-rich repeats are involved in the defense and stress responsiveness. Circadian is involved in circadian control.

(CsSWEET7a, CsSWEET7b, and CsSWEET15) exhibited relatively high transcript abundances in cucumber fruits (Fig. 6-G, H, and N). These results indicate that the cucumber *SWEET* genes are involved in various physiological and developmental processes.

3.6. Phylogenetic analysis of *SWEET* genes in cucumber, watermelon, and melon

Thus far, no *SWEET* genes have previously been identified and characterized in other Cucurbitaceae crops such as watermelon and melon. Using the same method as for the cucumber, the *SWEET* genes of watermelon were detected from the watermelon genome and resulted in the identification of 18 *ClaSWEET* genes. The 18 *ClaSWEET* genes were named according to their homologous genes in *Arabidopsis*. An unrooted phylogenetic tree was generated from the alignments of the 17 CsSWEET proteins and 18 *ClaSWEET* proteins, which resulted in the formation of four clades (clades I, II, III and IV) based on the classification by Chen *et al.* (2010) (Fig. 7). Each CsSWEET protein (with the exception of CsSWEET9 and CsSWEET17b) was clustered with a single watermelon *ClaSWEET* protein, and their similarity levels were especially high (bootstrap values ≥ 98%) (Fig. 7). *ClaSWEET*4a, 4b, and 4c were the three watermelon proteins closest to each other, and no homologs were found in the cucumber (Fig. 7). Furthermore, 18 putative melon *CmSWEET* genes were identified from the melon genome. The phylogenetic tree showed that each CsSWEET protein was clustered with a single melon *CmSWEET* protein except for CsSWEET9 which clustered with two, and their similarity levels were especially high (bootstrap values ≥ 95%) (Appendix D).

4. Discussion

The *SWEET* gene family is widely distributed in the plant kingdom and participates in many significant physiological and bio-

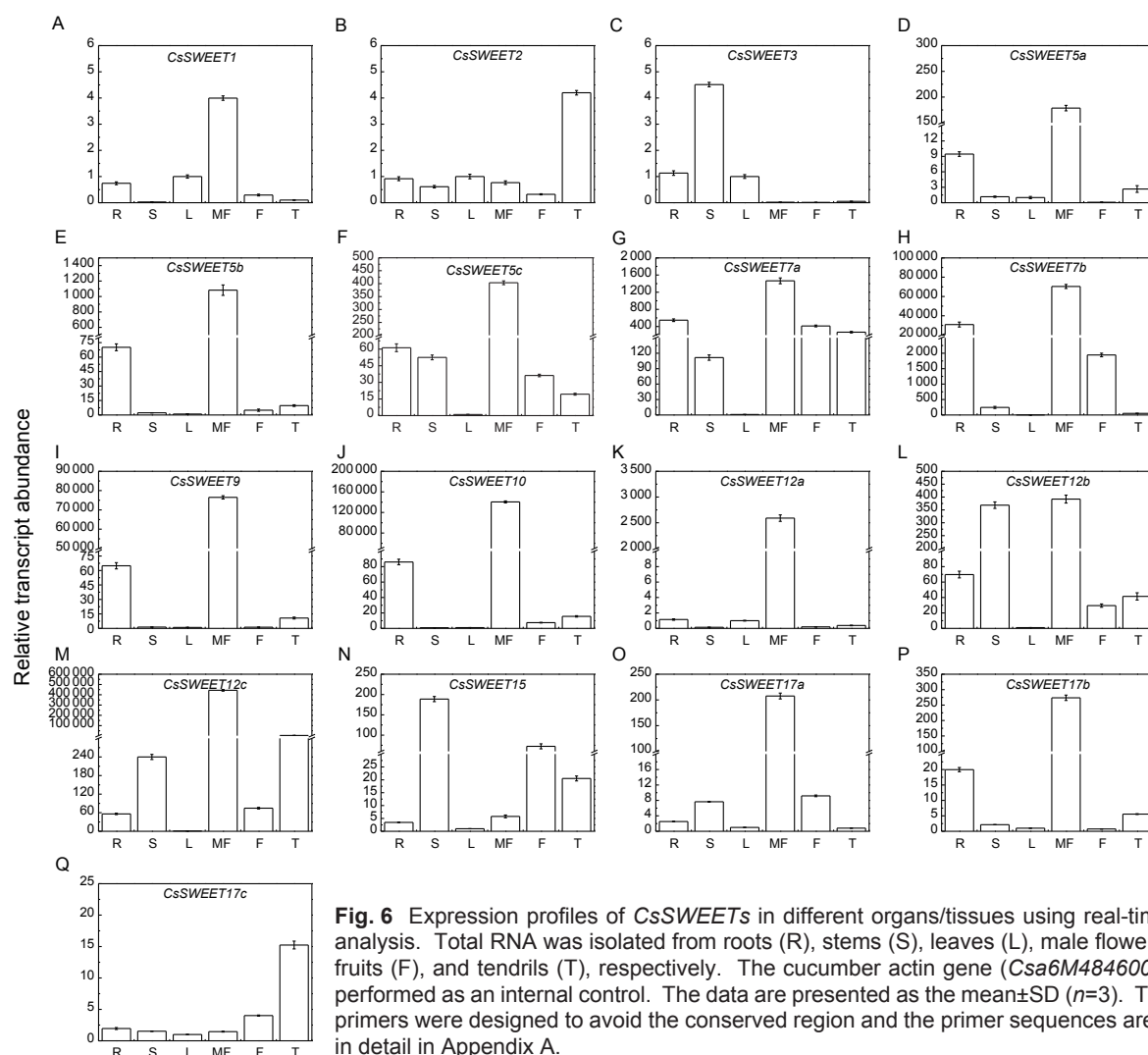


Fig. 6 Expression profiles of *CsSWEET*s in different organs/tissues using real-time PCR analysis. Total RNA was isolated from roots (R), stems (S), leaves (L), male flowers (MF), fruits (F), and tendrils (T), respectively. The cucumber actin gene (*Csa6M484600.1*) was performed as an internal control. The data are presented as the mean $\pm$ SD ($n=3$). The PCR primers were designed to avoid the conserved region and the primer sequences are shown in detail in Appendix A.

chemical processes (Yuan and Wang 2013; Chen 2014; Chandran 2015; Chen *et al.* 2015a; Eom *et al.* 2015; Patil *et al.* 2015). However, only a few species, such as *Arabidopsis*, rice, tomato, and soybean, of the *SWEET* gene family have been identified and characterized at the genome level (Chen *et al.* 2010; Yuan *et al.* 2014; Feng *et al.* 2015; Patil *et al.* 2015). To our knowledge, no systematic investigations on the *SWEET* gene family of Cucurbitaceae crops have been reported to date. In this study, we explored 17 *SWEET* genes in cucumber *via* an analysis of their phylogeny, chromosome distribution, gene structure, domain architecture, *cis*-acting regulatory elements, and expression profiles.

Recent studies have suggested that the *SWEET* gene family expanded during the course of evolution (Yuan *et al.* 2014; Patil *et al.* 2015). It has been proposed that gene duplications, including segmental, tandem, and whole genome duplications, play crucial roles in the evolution of various organisms (Xu *et al.* 2012). However, whole genome duplication, which is common in angiosperm plants and is an

important source of raw materials for gene genesis, is absent in the cucumber (Huang *et al.* 2009). In this study, two pairs of *CsSWEET* genes (*CsSWEET5b* and *CsSWEET5c*; *CsSWEET12b* and *CsSWEET12c*) were considered as tandem duplications (Fig. 1). Tandem duplication events also have been found in *SWEET* genes of soybean and rice (Patil *et al.* 2015). The coding sequences of the 17 *CsSWEET* genes were relatively conserved (Fig. 3). Conserved domains of the *SWEET* protein may be necessary for their key roles in plant life. Furthermore, two possible serine phosphorylation sites were found on the inner sides of the membrane area of *CsSWEET* proteins (Fig. 3). This indicates that the inner sides of the membrane area of the *CsSWEET* proteins are likely to be their important functional areas or their active regulatory regions, and their regulation is likely subjected to reversible phosphorylation/dephosphorylation, which is controlled by a protein phosphatase/kinase. These findings may represent a breakthrough in the analysis of the function of the cucumber *SWEET* gene family.

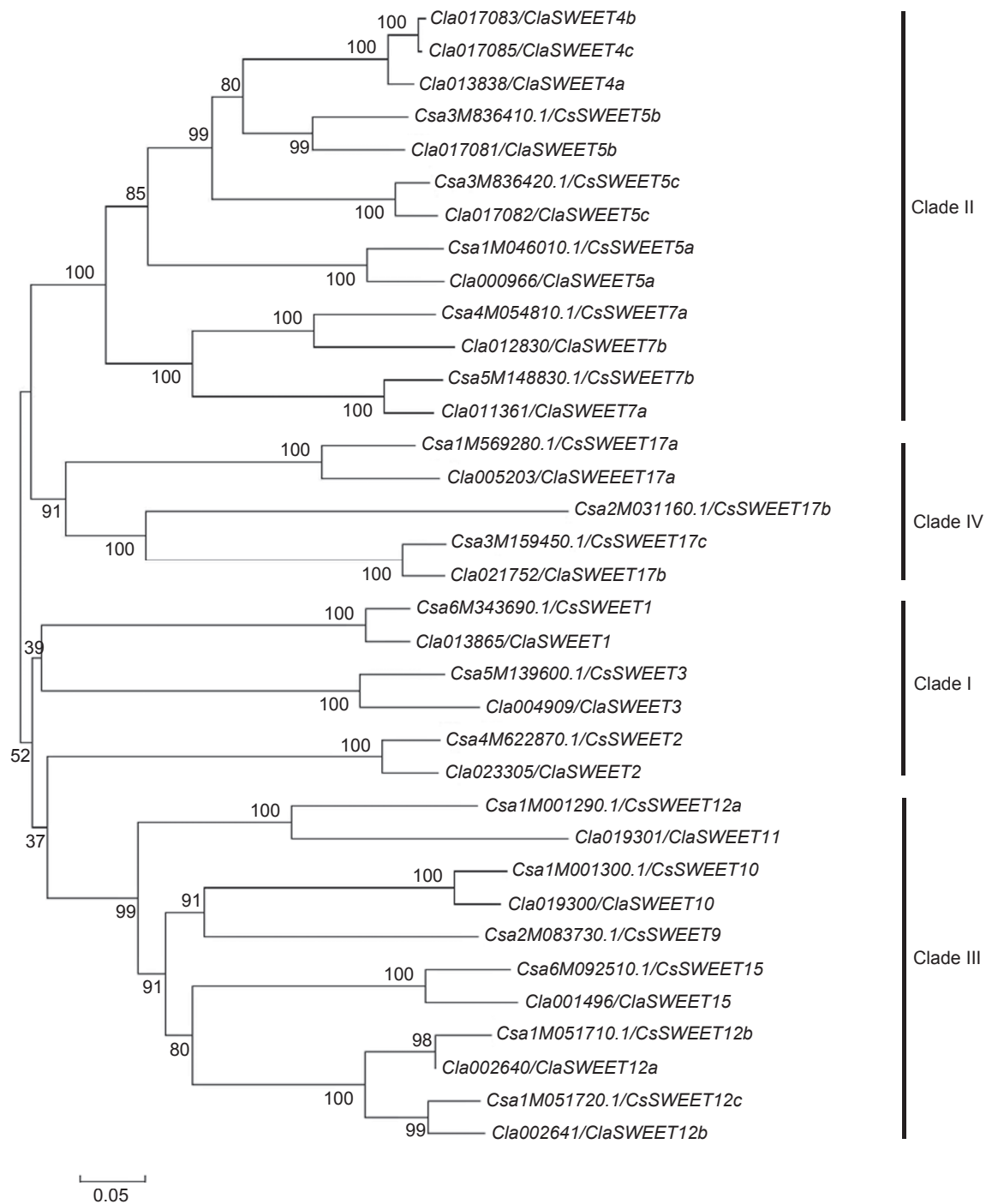


Fig. 7 Phylogenetic relationships of SWEETs genes in cucumber and watermelon. SWEETs genes fall into four clades based on the classification by Chen et al. (2010). Cs, cucumber; Cla, watermelon.

The *cis*-elements for plant hormone and stress-related responses in the promoter sequences of the 17 *CsSWEET* genes were also investigated. Except for *CsSWEET7*, at least one *cis*-element for the plant hormone response was discovered in the promoter regions of the 17 *CsSWEET* genes, implying that the corresponding hormones may have

an important role in their regulation (Table 2). Furthermore, all *CsSWEET* genes contain at least one *cis*-element for the stress-related response (Table 2), suggesting that these genes also play important regulatory roles in the stress response. However, further studies are needed to determine whether and how these *cis*-elements function in

cucumber plants.

As mentioned above, the *SWEET* genes play diverse functional roles during plant development. Although the roles of cucumber *SWEET* genes in these processes are not known, the possibility cannot be ruled out that cucumber *SWEET* genes, like the *SWEET* genes of *Arabidopsis* and rice, may be involved in these developmental processes. To date, most of the reported *SWEET* genes from *Arabidopsis* and rice have been associated with reproductive tissue development (Chu et al. 2006; Guan et al. 2008; Antony et al. 2010; Chen et al. 2010; Yuan and Wang 2013; Yuan et al. 2014; Chen L et al. 2015b). The results presented in this paper provide further evidence to support this observation; that is, most (70.58%) of the cucumber *SWEET* genes exhibited higher expression levels in male flowers than in other organs (Fig. 6). These results imply that cucumber *SWEET* genes may also be tightly associated with reproductive development. Different paralogs of this gene family may be specifically involved in different developmental stages or may be functionally redundant. Therefore, further detailed analyses are required to explore the temporal and spatial expression profiles and the biological functions of these genes during male flower development.

In many higher plants, such as *Arabidopsis* and rice, sucrose is the main form of carbohydrate transported in the phloem and distributed to various sinks. Phloem loading is the first important step in the long-distance transport of photoassimilate from the leaves to sink organs (Rennie and Turgeon 2009). In *Arabidopsis*, phloem loading involves an apoplastic step, and this process is driven by *AtSWEET11* and *AtSWEET12* localized to the plasma membrane of the phloem parenchyma of leaves (Chen et al. 2012). Except for sucrose-transporting plants, in certain species, such as Cucurbitaceae and Lamiaceae, raffinose family oligosaccharides (RFOs) are the primary translocated carbohydrates (Bachmann et al. 1994; Hu et al. 2009). In RFO-transporting plants, phloem loading follows an extensive symplasmic pathway, which is known as the polymer trap (Rennie and Turgeon 2009). According to the polymer trap mechanism, sucrose diffuses from the mesophyll symplasm into the intermediary cells (specialized companion cells) in the minor veins and is converted to RFOs, which are primarily the trisaccharide, raffinose, and the tetrasaccharide, stachyose. These RFOs are larger than sucrose and are apparently unable to diffuse back to the mesophyll through the intermediary cell plasmodesmata (Rennie and Turgeon 2009; Yadav et al. 2015). Obviously, this process does not require the participation of sucrose transporters, including *SWEET*. However, *CsSWEET1*, 2, and 3 exhibited a relatively higher expression in cucumber leaves (Fig. 6-A, B, and C). What are their physiological functions? Notably, recent studies have shown that *AtSWEET2* is the principal factor

that determines glucose contents in the tonoplasts of *Arabidopsis* leaves and roots (Chen et al. 2015). In addition, *CsSWEET1*, 2, and 3 all belong to clade I (Fig. 4) and are predicted as monosaccharide efflux transporters (Chen et al. 2015a). Therefore, *CsSWEET1*, 2, and 3 may not directly participate in phloem loading but instead are involved in regulating the monosaccharide contents in cucumber leaves, which significantly impacts carbon allocation and transport flux to the heterotrophic tissues of the plant. However, their specific functions deserve further study.

Although RFOs are the primary sugars transported in the cucumber phloem, the major sugars in the peduncles and fruits are sucrose and hexoses (glucose and fructose), which suggests that sucrose rather than RFOs is translocated into the fruit (Gross and Pharr 1982; Handley et al. 1983; Hu et al. 2009). Phloem unloading for sucrose in cucumber fruit follows an apoplastic pathway (Hu et al. 2011), and the subsequent hydrolysis to hexoses occurs via cell wall invertase (*CscwINV3* and *CscwINV5*). The hexose transporter (*CsHT3*) is then responsible for hexose uptake from the fruit apoplasm into the fruit storage parenchyma cells (Cheng et al. 2015). However, there are still many unanswered questions about this process. For example, little are known about the key transporters that participate in the first step of exporting sucrose from sieve element-companion cell complexes into the apoplasmic space. Further, the key players involved in the transport of fructose derived from the cleavage of sucrose in the apoplast space remain unknown because *CsHT3* may transport glucose but not fructose (Cheng et al. 2015). Our study has shown that some *SWEET* family genes, such as *CsSWEET7a*, *CsSWEET7b*, and *CsSWEET15*, have relatively high transcript abundances in cucumber fruits (Fig. 6-G, H, and N), which suggests that they participate in regulating the unloading process. However, further studies are needed to determine whether and how these genes function during cucumber fruit development. It is thought that the elucidation of the functions of *SWEET* genes during cucumber fruit development will provide some answers to these questions because *SWEET* genes may transport not only sucrose and glucose but also fructose (Chen et al. 2010; Guo et al. 2014). Additionally, some *SWEET* genes are expressed in the fruits of tomato (Feng et al. 2015), sweet orange (*Citrus sinensis*) (Zheng et al. 2014), and apple (*Malus domestica*) (Wei et al. 2014), but their precise roles remain unclear.

Furthermore, 18 *ClSWEET* genes and 18 *CmSWEET* genes exhibiting a high degree of similarity with *CsSWEET* genes were identified in watermelon and melon (Fig. 7 and Appendix D). Cucumber, watermelon, and melon are closely related, although cucumber has 7 chromosomes, watermelon has 11, and melon has 12. Analysis of the detailed chromosome-to-chromosome relationships reveals

a high degree of complexity of chromosomal evolution and rearrangement among these three important crop species of the Cucurbitaceae family (Huang *et al.* 2009; Garcia-Mas *et al.* 2012; Guo *et al.* 2013). Thus, it is not surprising that the similarity levels of *SWEET* genes in cucumber, watermelon, and melon are especially high (Fig. 7 and Appendix D), and these homologous *SWEET* genes may have similar functions.

5. Conclusion

In the present study, we performed a comprehensive genome-wide analysis of 17 cucumber *SWEET* genes with regard to their characteristics, exon-intron structures, genome distribution, conserved motifs, phylogeny, *cis*-element prediction, and expression analysis. Additionally, 18 putative watermelon *Cl**SWEET* genes and 18 melon *CmSWEET* genes were identified to provide further information about the *SWEET* gene families in cucurbits. The results from this study provide a basic understanding of the *CsSWEET* genes and may also facilitate future research that may elucidate the detailed roles of *SWEET* genes in cucumber and other Cucurbitaceae crops.

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Appendix associated with this paper can be available on <http://www.ChinaAgriSci.com/V2/En/appendix.htm>

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